Government Subsidy & Grant Administration Platform

From application to impact — a transparent, trackable subsidy lifecycle.

TRANSPARENT | EFFICIENT | ACCOUNTABLE

SAGAR • Infosys Internship Project

Team: Mahesh | Kasturi | Akshaya



One platform.
A clearer journey for every beneficiary.

PROBLEM STATEMENT

The idea began with one citizen’s difficult journey



Meet Savitri — a woman entrepreneur in a rural community

She hears about a women’s entrepreneurship subsidy that could help her start a small business.

01	**Unclear eligibility**	She is unsure whether her age, income and business plan meet the scheme rules.
02	**Paperwork and repeat visits**	Documents are submitted through a process with little visibility into what is missing.
03	**No clear status**	She cannot easily tell who is reviewing the application or when she will hear back.
04	**Trust is weakened**	Delays and fragmented records make the process feel difficult to understand.

A fictional, composite scenario inspired by common process challenges — not a real named beneficiary.

Our guiding question: How can we make subsidy access easier to follow, verify and trust?

Problem statement: fragmented processes create avoidable friction

Manual eligibility checks

- Repeated review of scheme rules
- Higher chance of inconsistent decisions

Scattered verification records

- Field and administrative evidence can be fragmented
- Harder to trace decisions end-to-end

Limited fund visibility

- Difficult to track staged releases and compliance
- Less insight into scheme / regional utilization

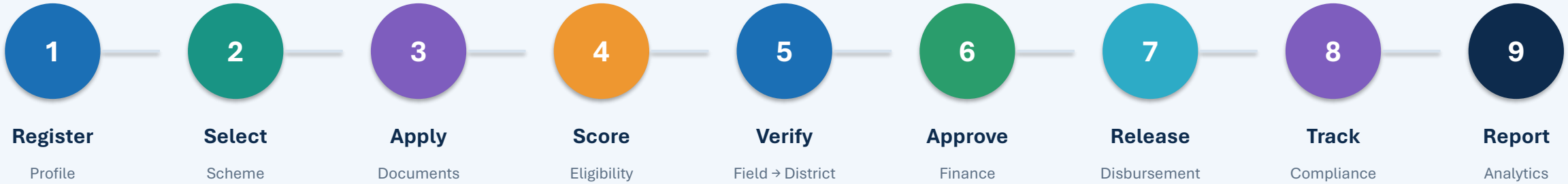
The core problem

Traditional subsidy workflows can be slow, hard to audit and difficult for applicants to understand — from initial eligibility through final fund utilization.

Project objective: centralize the lifecycle, apply scheme-specific eligibility rules, route decisions through authorized roles, and connect disbursement to compliance evidence.

Our solution: one connected subsidy lifecycle

SAGAR brings the beneficiary journey, officer decisions, fund release and reporting into a single workflow.



C For beneficiaries

Clear scheme criteria, document submission, score feedback and application tracking.

O For officers

Role-specific queues for field checks, district review, finance approval and disbursement.

A For administrators

Audit visibility, utilization analytics, pending milestones and downloadable reporting.

Project journey: all four milestones completed

MILESTONE 1

Weeks 1–2

Foundation & setup

Environment, database schema, workflow design

MILESTONE 2

Weeks 3–4

Eligibility & verification

Scheme rules, 100-point scoring, multi-role workflow

MILESTONE 3

Weeks 5–6

Disbursement & compliance

Plans, milestone tracking, reminders, analytics

MILESTONE 4

Weeks 7–8

Security & integration

Role access, audit logging, REST APIs, testing/docs

Outcome: the four milestone workstreams are integrated into one application lifecycle.

Milestone 1 — foundation, environment and process design

1 Environment & project setup

- Java 21 and Spring Boot project foundation
- Maven Wrapper for repeatable local builds
- MySQL connectivity and JPA / Hibernate persistence
- Configuration for application startup

2 Process & data design

- Beneficiary and scheme data models
- Eligibility criteria and score architecture
- Verification status and role flow
- Regional budget fields and disbursement lifecycle

✓ Foundation delivered

- Core entities and repositories established
- Web / controller / service / persistence layers
- Initial API and integration structure
- Base login, registration and dashboard views

Milestone result: a working technical foundation on which eligibility, workflow and reporting features could be built.

Milestone 2 — beneficiary, scheme and workflow capabilities

A Beneficiary & scheme master

- Beneficiary registration and profile management
- Scheme details and configurable criteria
- Seven scheme configurations loaded at startup
- Required and optional document definitions

B Eligibility engine

- 100-point scoring: profile + mandatory docs + optional docs
- Scheme-specific age / income and other gates
- Live score and final submission validation
- Missing-document and criterion-level feedback

C Multi-level workflow

- Field officer ground verification
- District officer review and send-back / rejection
- Finance approval and sanction amount
- Cooling period, applicant re-verification and resubmission

Milestone result: scheme-aware eligibility and sequential role-based decision queues.

Eligibility engine — explainable, scheme-aware scoring

100-POINT SCORE



50 pts
Profile

40 pts
Mandatory docs

10

Profile factors include age, income, government ID, occupation, gender, disability and residence.

All scheme gates + mandatory evidence must pass.

Live scoring

Uses the same server-side scoring engine to show score, eligibility flag and missing documents before submission.

Submission validation

Application record is created only after eligibility checks and mandatory document validation pass.

Applicant feedback

- Criterion-by-criterion pass / fail
- Total score and uploaded-document points
- Missing mandatory documents and status

Correction pathway

- Send-back / rejection handling
- 30-day cooling period when applied
- Resubmission recalculates score

Gate checks: scheme age range • income ceiling • government ID • configured occupation / gender / disability / residence rules • all mandatory documents. A failed gate or missing mandatory file returns the user to the form; no incomplete application is silently created.

Milestone 3 — staged disbursement and compliance

$ Disbursement plans

- Plan created for an application
- Milestone-wise amount breakdown
- Due dates and plan status
- Actual release tracked against plan

! Compliance monitoring

- Document / utilization evidence tracking
- Due-date calculations
- Reminder history and automated reminders
- Overdue and non-compliance flags

✓ Controlled fund release

- Finance approval routes to disbursement stage
- Dedicated disbursement officer queue
- UTR number captured for release
- Disbursed timestamp and remarks retained

1 Approved — 2 Plan set — 3 Evidence due — 4 Funds released

Cooling period: rejected / send-back cases may enter COOLING_PERIOD; coolingPeriodEndDate blocks immediate resubmission for 30 days. After expiry, evidence can be updated, eligibility is recalculated, and resubmissionCount increases. If still ineligible, APPLICANT_REVERIFICATION remains active.

Milestone 4 — analytics, integration and operational controls

Analytics & reports

- Scheme-wise and region-wise summaries
- Sanctioned vs released fund visibility
- Pending / overdue milestone analysis

Security & roles

- Spring Security and role-based access
- BCrypt password hashing
- Role-specific officer queues and jurisdiction checks

Audit trail

- Records actor, role, action and entity
- Captures previous / new values and remarks
- Recent audit logs and entity history

REST API support

- Versioned application tracking endpoints
- Scheme and officer API controllers
- Authenticated users can retrieve their own application data

Testing, documentation & deployment readiness

- Project configuration and build structure
- Basic application context test included
- Deployment / external treasury integration should be validated in target environment

Implementation note: API endpoints and audit/security foundations are present; external production-system connectivity depends on environment configuration.

System architecture — layered, maintainable and role-aware

PRESENTATION	Thymeleaf templates • citizen / officer / admin dashboards
CONTROLLER	Web, eligibility, beneficiary, admin and milestone controllers
SERVICE / RULES	EligibilityService • SchemeService • BeneficiaryService • disbursement services
PERSISTENCE	Spring Data JPA repositories • users • beneficiaries • schemes • applications • documents
INFRASTRUCTURE	Java 21 • Spring Boot • Spring Security • Hibernate • MySQL • Maven

SECURITY BOUNDARY

Public

Portal, login, registration, scheme details

Authenticated

Dashboards and application actions

Role-specific

Field, district, finance, disbursement and admin

Separation of concerns makes workflow logic easier to test, maintain and extend.

Core data models and scheme master configuration

P Beneficiary model

- Government ID and contact details
- Age, income, occupation, gender
- Disability and residence attributes
- User relationship and profile data

S Scheme model

- Age / income thresholds
- Occupation / gender / disability / residence gates
- Required and optional evidence
- Regional budget fields

A Application & workflow records

- Eligibility status and score
- Sanction amount and application date
- Uploaded document records and remarks
- Status, re-submission and disbursement data

Seven configured schemes (from the current project master data)

Scheme	Age range	Income limit	Additional gate
Education Subsidy	18–30	≤ ₹5L	Student
Farmer Support Grant	18–65	≤ ₹4L	Farmer
Housing Assistance	21–70	≤ ₹3L	—
Women Entrepreneurship	18–55	≤ ₹8L	Female
Senior Citizen Assistance	60–100	≤ ₹3L	—
Divyang Disability Support	18–80	≤ ₹3.5L	Disability = Yes
Rural Skill Development	18–45	≤ ₹4.5L	Rural residence

Additional scheme gates: Student • Farmer/Agriculturist • Female applicant • Disability = Yes • Rural residence. Evidence varies by scheme (e.g., Student ID, Land/KCC, Property, Business Plan, UDID or Rural Residence proof).

End-to-end workflow — from registration to accountable reporting

1
Register
Beneficiary profile

2
Select
Scheme criteria

3
Apply
Form + evidence

4
Score
100-point engine

5
Verify
Field → District

6
Approve
Finance review

7
Disburse
Plan + UTR

8
Track
Compliance

9
Report
Analytics

Status tracking, audit records and reporting connect each stage instead of leaving the process as isolated steps.

Governance built into the workflow: roles, audit and APIs

1 Beneficiary

Profile, application, score feedback and tracking

2 Field officer

Ground verification and first-level action

3 District officer

Eligibility review, remarks and send-back

4 Finance approver

Sanction decision and approval

5 Finance officer

Release control and UTR capture

6 Administrator

Officer management, monitoring and audit review

Audit logging

Actor + role + action + entity + before/after values + remarks.

REST APIs

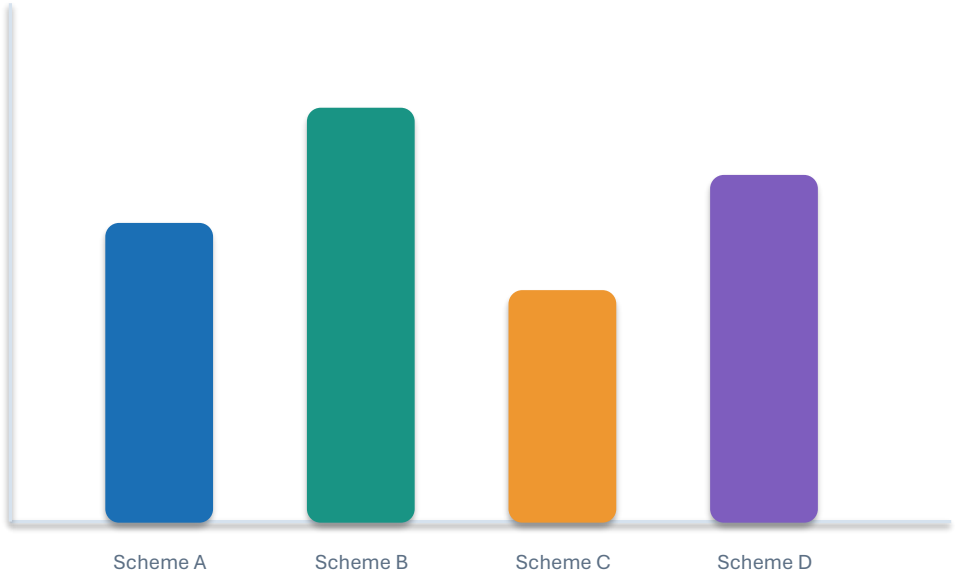
Versioned endpoints for scheme, officer and authenticated application tracking.

Security foundations include Spring Security, BCrypt, role checks and jurisdiction-aware filtering.

Fund utilization analytics — move from records to decisions

Analytics view (illustrative layout)

Actual dashboard values are calculated from stored applications and disbursement plans.



Budget and release

- Allocated budget vs actual release
- Remaining budget responds to released funds

Compliance and turnaround

- Pending / overdue milestones
- Approval turnaround and non-compliance signals

Reporting outputs

- Scheme and region summaries
- PDF / Excel downloads where supported by the application

Analytics is designed to support better oversight; chart bars above are placeholders, not project data.

What the implementation enables — and what comes next

Value delivered

- Centralized beneficiary and scheme information
- Consistent, explainable eligibility scoring
- Sequential verification with send-back / resubmission
- Staged disbursement and compliance monitoring
- Audit visibility, role-based access and tracking APIs
- Scheme / region analytics and report downloads

Recommended next steps

- Run deeper automated tests for scoring and workflow edge cases
- Connect and validate real treasury / beneficiary systems
- Complete deployment hardening in the target environment
- Review access, privacy, backup and monitoring policies
- Validate report outputs and analytics with real operational data
- Gather citizen / officer feedback for usability improvements

Our outcome: a working foundation for transparent, traceable and compliance-linked subsidy administration.

The people behind SAGAR

M

Mahesh

FRONTEND & UI

Built the citizen-facing and officer-facing screens; worked on registration, eligibility forms, dashboards, result presentation and UI flow.

K

Kasturi

ELIGIBILITY & SCHEME RULES

Defined scheme-specific criteria and scoring requirements; contributed to scheme master configuration and eligibility presentation.

A

Akshaya

BENEFICIARY & WORKFLOW

Worked on beneficiary data and profile flow; contributed to application lifecycle, verification routing, resubmission and workflow coordination.

Team integration: frontend captures inputs → eligibility engine evaluates → workflow services route decisions → dashboards expose status to authorized users.

Thank You!

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